

4.6 relative velocity and acceleration

$\vec{v}_{BA}$ means the velocity of B observed by A.

$$\vec{v}_{AB} = -\vec{v}_{BA}$$

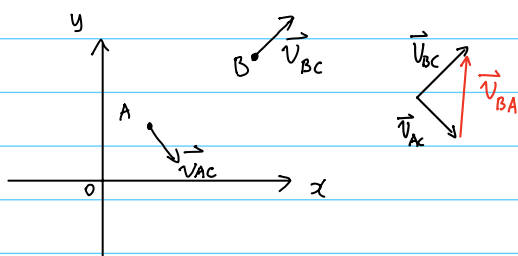
Galilean transformation:
$$\vec{v}_{CA} = \vec{v}_{CB} + \vec{v}_{BA} \quad (1)$$

• consistent with the "head to tail" rule

e.g. Given $\vec{v}_A$, $\vec{v}_B$, find velocity of B observed by A.

Let A, B be two particles, C be the origin. $\vec{v}_A$ means $\vec{v}_{AC}$, $\vec{v}_B$ is $\vec{v}_{BC}$

$$\begin{aligned} \vec{v}_{BA} &= \vec{v}_{BC} + \vec{v}_{CA} \\ &= \vec{v}_{BC} - \vec{v}_{AC} = \vec{v}_B - \vec{v}_A \end{aligned}$$



e.g. if $\vec{v}_A = \hat{i}$, $\vec{v}_B = 0$, then $\vec{v}_{BA} = -\hat{i}$

• Now let A, B be two reference frames. C is the object being observed.

Take the time derivative for (1):

$$\frac{d\vec{v}_{CA}}{dt} = \frac{d\vec{v}_{CB}}{dt} + \frac{d\vec{v}_{BA}}{dt} \Rightarrow \text{If } \vec{v}_{BA} = \text{constant, } \vec{a}_{CA} = \vec{a}_{CB}$$

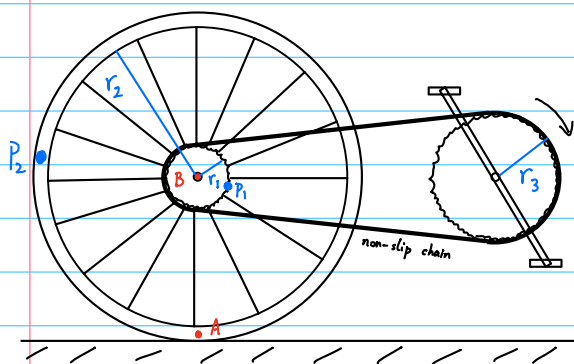
That means if two reference frames move with a constant relative velocity, the acceleration observed from these two reference frames are equal.

e.g. $\vec{a}$ of a free-falling ball is the same measured from ground and from a moving train with a constant velocity.

exercise: Riding a bicycle.

$$r_2 = 4 r_1, \quad r_3 = 2 r_1$$

Given T_3 , r_1 , find a_{c1} , a_{c2} , and speed of bicycle.



$$\omega_3 = \frac{2\pi}{T_3} \quad v_3 = r_3 \omega_3 = \frac{2\pi r_3}{T_3}$$

$$v_1 = v_3$$

$$a_{c1} = \frac{v_1^2}{r_1} = \frac{4\pi^2 r_3^2}{T_3^2 r_1} = \boxed{\frac{16\pi^2 r_1}{T_3^2}}$$

$$\omega_1 = \frac{v_1}{r_1} = \frac{2\pi r_3}{T_3 r_1} = \frac{4\pi}{T_3}$$

$$\omega_2 = \omega_1$$

$$a_{c2} = \omega_2^2 r_2 = \frac{16\pi^2}{T_3^2} r_2 = \boxed{\frac{64\pi^2 r_1}{T_3^2}}$$

check: 1) with the same ω , $a_c \propto r$,

$a_{c2} : a_{c1} = 4 : 1 = r_2 : r_1$, makes sense.

2) dimension analysis $\frac{16\pi^2 r_1}{T_3^2}$ unit is $\frac{m}{s^2}$ consistent with a .

$$v_2 = r_2 \omega_2 = 4 r_1 \cdot \frac{4\pi}{T_3} = \frac{16\pi r_1}{T_3}$$

speed of A observed by B. ↗

speed of bicycle = speed of B observed by A (ground).

$$= \left| \frac{-16\pi r_1}{T_3} \right| = \boxed{\frac{16\pi r_1}{T_3}}$$

Brief review

We learned how to describe motions using $\vec{r}(t)$, $\vec{v}(t)$, $\vec{a}(t)$

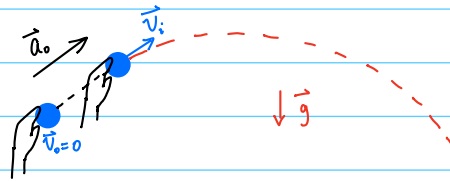
$$\left. \begin{aligned} v_f &= at + v_i \\ x &= \frac{1}{2}at^2 + v_i t + x_i \end{aligned} \right\} \text{kinematic equations for constant } a \text{ motion}$$

• one dimension

• two dimension $\xrightarrow{\text{decompose}}$ $\begin{cases} x \text{ direction} \\ y \text{ direction} \end{cases}$

mechanics $\begin{cases} \text{kinematics} : \text{describe the motion} \\ \text{dynamics} : \text{causes of change in motion} \end{cases} \rightarrow \text{forces}$
Newton's Laws

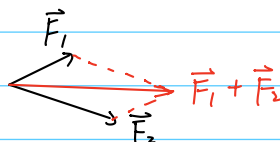
5-1 Force



throw a ball
my hand exerts a force on the ball, giving it an initial velocity.
the earth exerts a gravitational force on the ball.

contact force, field force
but microscopically both are field force

force is a vector, follows vector operation rules.



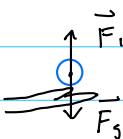
unit = N
(Newton)

5.2

Newton's first law of motion (law of inertia)

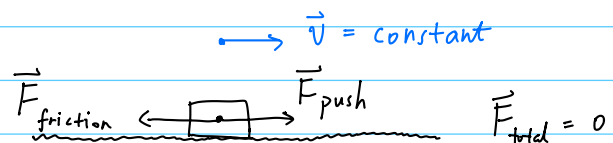
" An object which does not experience a net force
undergoes motion with constant velocity,
when measured from an inertial reference frame."

notes: 1) net force = sum of all force vectors
 exerted on this object.

e.g.1.  $\vec{F}_{\text{total}} = \vec{F}_1 + \vec{F}_2 = 0$

e.g.2. (seems counter intuitive)

I am pushing a block on a rough surface

 $\vec{F}_{\text{friction}} \leftarrow \text{block} \rightarrow \vec{F}_{\text{push}} \quad \vec{F}_{\text{total}} = 0$

In reality, friction is everywhere.

For this reason, in our daily life, it is impossible to see something moving in constant speed without applying a force.

Before Newton (~1600's), people had the false knowledge that "force is required to maintain x the constant velocity".

e.g.3. in outer space, no gravity, no air friction, an object will keep moving in the same direction with same speed.

2) $\vec{v} = \text{constant}$ means constant speed along a straight line. This includes $\vec{v} = 0$ (at rest).

3) how reluctant an object is to change its velocity under a force is called "inertia".

4) This law only applies in "inertial frames":
a stationary or uniformly moving frame of reference.

e.g.



Are these inertial frames?

track ✓

train with constant velocity ✓

child on train walking smoothly ✓

the ball in the air threw by the child ✗

another train that is slowing down ✗

the axis of the wheel ✓

a point on the edge of the wheel ✗

e.g.2. for motions on earth surface, the ground is a good enough inertial frame.

For satellite, the center of earth is a good enough inertial frame, but for the earth's revolution around the sun, it is not.

5) This law says that in an inertia frame,

$$\text{If } \vec{F}_{\text{total}} = 0, \quad \vec{a} = 0$$

What happens to an object if $\vec{F}_{\text{total}} \neq 0$?

5.4 Newton's second law

When viewed from an inertial reference frame, the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass:

$$\vec{a} = \frac{\vec{F}}{m} \quad \text{or} \quad \boxed{\vec{F} = m \vec{a}} \quad \text{here } \vec{F} \text{ is net force.}$$

Force causes $\vec{a}$ with a proportional constant called mass

symbol : m

SI unit : kg

For the same $\vec{F}$, bigger m smaller $\vec{a}$
smaller m bigger $\vec{a}$

Definition of Newton : $1 \text{ N} \equiv 1 \text{ kg} \cdot \text{m/s}^2$

- Newton's 1st law is a special case of the 2nd law.

e.g. Gravitational force and free fall motion

For a ball during free fall, the only force experienced by the ball is gravity.

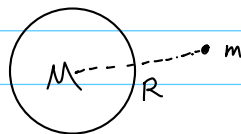

$$\vec{F}_g = m\vec{a} = m\vec{g}$$

$\vec{F}_g$ is exerted by $\underset{\Delta}{\text{Earth}}$ on the $\underset{\Delta}{\text{ball}}$.

gravitational force
(in general)

$$F_g = \left[\frac{GM}{R^2} \right] m$$

gravitational acceleration g



For different planet or different distance, g is different.

$g \approx 9.8 \text{ m/s}^2$ only for Earth surface.

5-6

Newton's Third Law ("action-reaction")

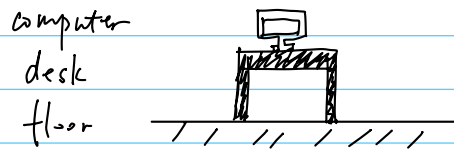
$$\vec{F}_{12} = -\vec{F}_{21}$$

exerted by object 1 on object 2

exerted by object 2 on object 1

equal in magnitude
opposite in direction

e.g.



Action-reaction pairs :

{ normal force exerted by computer on desk
normal force exerted by desk on computer

{ normal force exerted by desk on floor
normal force exerted by floor on computer

{ gravity force exerted by Earth on computer
gravity force exerted by computer on Earth

{ gravity force exerted by Earth on desk
gravity force exerted by desk on Earth

opposite object/subject.

"normal" means perpendicular.

normal force is when two objects pressing on each other.

This pressing force is always normal to the interface.

Problem analysis steps :

① "Free body diagram"

isolate one object and all the forces exerted on it.

e.g. the desk



F_1 : normal force from floor

F_2 : normal force from computer

F_3 : gravity force of desk

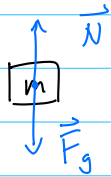
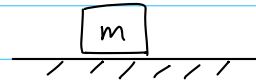
② sum all forces and use Newton's second law

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = m\vec{a} = 0 \quad (\text{desk is stationary})$$

Example 1

block on ground

1) analyze the forces.



$\vec{N}$ = normal force from ground (earth)

$\vec{F}_g$ = gravity force from earth

$$F_g = mg$$



$\vec{N}'$ = normal force from block

$\vec{F}_g'$ = gravity from the block

(yes, the mass of the block also pulls the earth)

2) Find the action-reaction pairs?

$\vec{N}$ & $\vec{N}'$

$\vec{F}_g$ & $\vec{F}_g'$

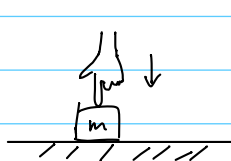
* $\vec{N}$ and $\vec{F}_g$ are not action-reaction!

3) Given m and g , find $\vec{N}$

$$\vec{N} + \vec{F}_g = 0 \quad (\text{block is at rest})$$

$$\vec{N} = -\vec{F}_g \quad \left\{ \begin{array}{l} |\vec{N}| = |-\vec{F}_g| = mg \\ \text{direction of } \vec{N} \text{ is opposite to } \vec{F}_g, \text{ i.e., vertically upward.} \end{array} \right.$$

4) If I press on the block, how will N change?



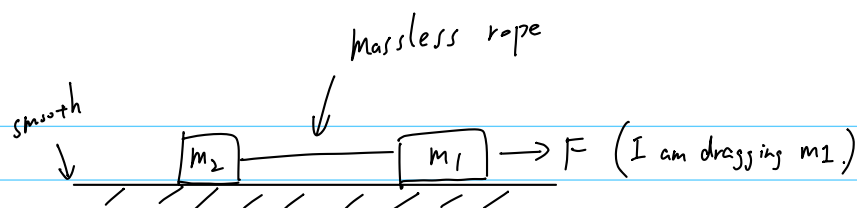
$$\vec{N} + \vec{F}_g + \vec{F} = 0$$

$$\vec{N} = -(\vec{F}_g + \vec{F})$$

$$|N| = |F_g| + |F| > |F_g|$$

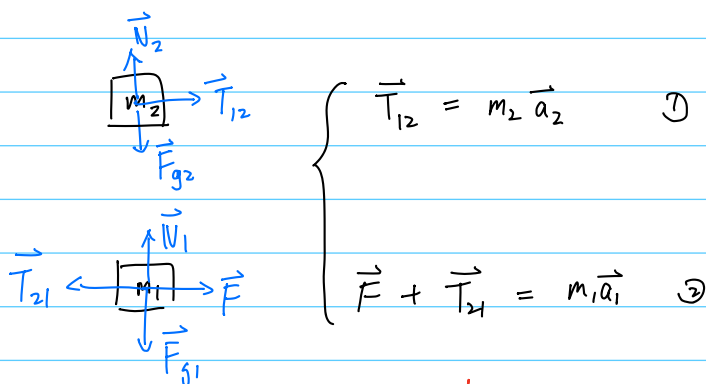
N will increase.

Example 2



Two blocks accelerate together on a smooth surface.

Given $F = 10\text{ N}$, $m_1 = 2\text{ kg}$, $m_2 = 3\text{ kg}$, how much is tension force in rope?



Set up coordinates.



Let $\vec{a}_2 = a \hat{i}$, then $\vec{a}_1 = \vec{a}_2 = a \hat{i}$ (accelerate together)

Let $\vec{T}_{12} = T \hat{i}$, then $\vec{T}_{21} = -\vec{T}_{12} = -T \hat{i}$ (Newton's 3rd law)

① and ② become:

$$\begin{cases} T \hat{i} = m_2 a \hat{i} \\ F \hat{i} - T \hat{i} = m_1 a \hat{i} \end{cases} \xrightarrow{\text{divide } \hat{i} \text{ from each term.}} \begin{cases} T = m_2 a \\ F - T = m_1 a \end{cases}$$

a, T unknown.

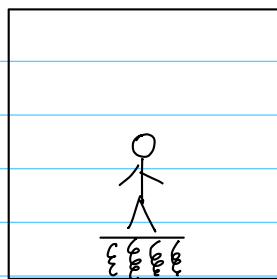
Replace a to find T :

$$F - T = m_1 \cdot \frac{T}{m_2}$$

$$F = \left(\frac{m_1}{m_2} + 1 \right) T$$

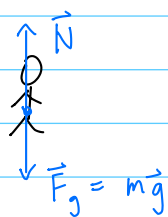
$$T = \frac{m_2}{m_1 + m_2} F = \frac{3}{2 + 3} \times 10\text{ N} = \boxed{6\text{ N}}$$

example 3



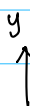
person standing on a scale in an elevator.
What does the scale read if the elevator is accelerating upward/downwards or moving with constant speed?

Let's say the whole system is moving with $\vec{a}$ (can be either upward or downward).



$$\Sigma \vec{F}_y = \vec{N} + m\vec{g} = m\vec{a}$$

$$\vec{N} = m\vec{a} - m\vec{g} \quad (\text{Scale reading is } |\vec{N}|)$$



$$\vec{g} = -g, \quad \text{Let } \vec{a} = a\hat{j}, \quad N = ma + mg$$

If not accelerating, $a=0$, $N=mg$, scale reading = body weight at rest

If accelerating upwards, $a>0$, $N>mg$, scale reading > body weight at rest
($\vec{v}$ up & v increasing or $\vec{v}$ down & v decreasing.)

If accelerating downwards, $a<0$, $N<mg$, scale reading < body weight at rest
($\vec{v}$ up & v decreasing or $\vec{v}$ down & v increasing.)

If $a = -g$, $N=0$, scale reading = 0, "appears weightless"
(elevator in free fall)

* What if $a < -g$? (accelerate downward with a larger acceleration than g)
Person will hit their head to the ceiling T.T.

* Imagine when elevator start at rest from 1st Fl, accelerate up till a constant speed, then slow down to be at rest at 3rd Fl,

